Evaluation of Connectives Acquisition in a Humanoid Robot Using Direct Physical Feedback

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Abstract. In this paper, we propose a method where humanoid robot acquires meanings of grammatical connectives using direct physical feedback from human. Our system acquired 70% connectives of all connectives taught by subjects. It can be also said that robot partially learned the concept of time.

1 Introduction

Humanoid robots should interact with humans using natural language and work adaptively in the same environment as human's. However, it is too difficult to suppose every environmint preliminarily. Therefore, algorithms where a robot will adaptively learn concepts of actions represented by natural languages are required.

Thus if we assume that infants and robots having less knowledge were analogical, we can also assume that Inductive Learning[1][2] should be efficient for the progress of robot's language acquisition, because infants learn inductively[3]. Furthermore, according to studies of so called "Embodied Intelligence"[4], embodied systems can easily acquire intelligence by mutual interaction with their environment[5]. Therefore we assumed that direct physical feedback (see Chapter 3) is also effecient for this problem.

As the first step, we start to acquire connectives. In this reserch, our target language is Japanease, and we will use *italic* when giving a Japanese examples.

2 Element of Action and Connectives

In this paper, we define Element of Action (EoA) as follows. EoA contains only one verb. EoA shows final physical position of action. We also call a command that meant EoA an **EoA command**. The final position of EoA is decided by average of direct physical feedback, by robot's users.

We define connectives as grammatical expressions which join two EoA commands. We also call a command that contains connectives a **connective command**. According to our definition of connectives, the connectives can also contain verbs to connect two EoAs.

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3 Way of Teaching(Direct Physical Feedback)

A humanoid robot learns by being taught by human how to move its body. This becomes an universal and natural method which allows us to teach within an allowance of a humanoid robot's body structure. This approach helps such robot to reproduce movements with very high certainty.

4 System

4.1 Humanoid Robot

For our experiments, we use a humanoid robot KHR2-HV[6]. The robot is equipped with 17 motors, and it sends signals describing its states of motors only. It does not have any other sensors.

We use 7 motors of all motors which is upper half of the robot's body. The particular motors are abbreviated as follows. [H]:Head, [LS]:Left Shoulder, [LA]:Left Arm, [LH]:Left Hand, [RS]:Right Shoulder, [RA]:Right Arm, [RH]:Right Hand. All motors' 180 degrees movements were divided into 10 ranks, 18 degrees each.

4.2 System Flow

Our system algorithm is outlined below.

1. A user inputs a command in natural language (Japanese) from a keyboard.
2. The robot applies rules or uses examples that have been previously acquired, and performs an action.
3. If the robot performs a wrong action or does not perform any action, the user teaches it a proper action by direct physical feedback. And then the system adds a pair of command and action to the example database.
4. Then, the system generates abstracted rules from the example database by Inductive Learning process.

3 Representation of Action

action is represented as a matrix shown in Figure 1. While a user moves the ot's body, it measures degrees of all the angles of motors and digitalize them y second. If a motor does not move, degree of its angle is described as "x". do not use motor speed information yet and we plan to use it when acquiring obs.

Example

ample consists of two parts. One is a natural language command morpho- ly analyzed by MeCab[7], and another is an action matrix. We use mor- ic analysis to make learning more effective. The analyzer absorbs endings s which change variously depending on conjugation.

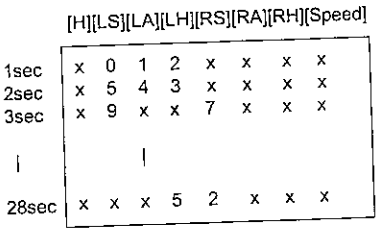


Fig. 1. Action Matrix

4.5 Generation of Rules by Inductive Learning

The system acquires meanings of connectives by generation of rules using Inductive Learning (Figure 2). The rule is consisted of natural language part and action matrix part. First, the system brings out three examples (example 1, 2 and 3 in Figure 2) from example database. Then, from natural language parts, the system realizes that example 1, 2 are EoA commands and example 3 is connective command. And the system parameterizes example 3 (@1 after that @2). Next, in action matrix parts, the system finds final positions of EoA commands (xxxx8x5, x9xxxxx) and parameterizes the motor positions of example 3. Finally, each rule has an adaptation value which is the number of identical rules.

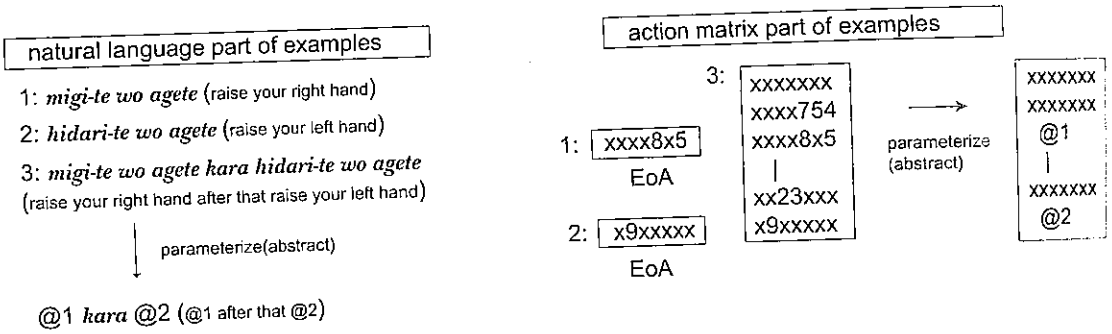


Fig. 2. Generation of Rules by Inductive Learning

5 Experiment

We implemented the system and experimented on learning connectives. We made 4 subjects (age 20-30, all were male students of graduate school majoring math and science) input commands and teach actions to the robot. We got 157 connective commands (all 157 commands differed) and 25 connectives (Table 1). We restricted the number of EoA commands to 10 for performing the experiment.

The subjects evaluate the system responses using following evaluation marks for every system output: Correct Response, Semi-correct Response, Erroneous Response, No Response. We define Correct Response and Semi-correct Response as proper responses. In case of Erroneous Response and No Response, the subjects have to teach the correct action to the robot.

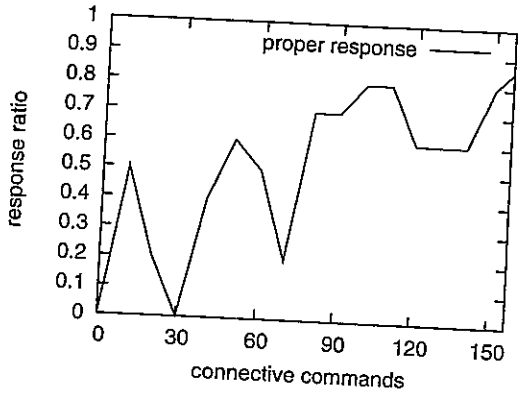


Fig. 3. Shift of the system response (connective commands)

Figure 3 shows changes in proper response ratio per 10 commands of all responses to the 157 connectives commands. This shows that the effective response line becomes stable from the third person (from the 75 connective command). Though the system did not have enough knowledge to reply with effective responses in the beginning, it managed to correct its knowledge enough to start reacting properly.

Table 1. connectives from subjects

kara (and then) **	*nagara* (and at the same time) **
mae-ni (before) **	*ato* (after) **
nai-de (without doing) **	*None* (None) **
to-doji-ni (while) **	*kara-yukkuri* (before slowly) *
sugu (do just before) *	*to-mise-kakete* (pretend to) *
ato-de (after that)	*sukoshi-matu-kara* (wait a minute)
ato-de-takusan-matsu (wait some minutes)	*tsutsu* (and)
shibaraku-shite-kara (and wait a minute)	*sonomama* (keep on)
ato-de (after that)	*soshite* (after)
soshite-takusan-matsu-kara (wait couple of minutes)	*toki* (when)
kara-sugu (soon before)	*sukoshi-matsu-kara* (wait a second)
takusan-matsu-kara (wait a couple of seconds)	

We evaluated rules applied 3 or more times in learning experiment. There are 10 such rules obtained (rules marked ** or * in Table 1). Other 4 subjects evaluated actions which the system outputs applying the rules, and 7 of 10 connectives (rules marked **) were evaluated as "successful" or "almost successful" by all subjects. And 3 connectives (rules marked *) were evaluated as "erroneous" by at least one subject.

As a result, we can say that the system can acquire connectives which signify simple relations of order. For example, "*nagara* (and at the same time)" or "*mae-ni* (before)" or "*naide* (without doing)". However, the system could not acquire 3 connectives. Firstly, "*sugu* (do just before)" and "*kara-yukkuri* (before

slowly)" had problems with acquiring proper timing. This can be acquired by increasing feedback. Secondly, "*to-mise-kakete* (pretend to do)" was incorrect due to difficult verb existence inside the connective. The system still can not understand all verbs correctly.

6 Conclusions

We proposed a method for grammatical connectives acquisition. We implemented the system, and performed learning and evaluation experiments.

From the results, we understood that learning connectives by teaching with direct physical feedback gives precision of connectives acquisition 0.7. It can be also said that our system was able to acquire a partial concept of time by embodying it. We also discovered that our system could not acquire some connectives due to the lack of understanding of verbs meaning. We try to develop mechanism of understanding verbs in the next step.

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